Chen 2020 NPC — analysis summary

RUN 1 OF 7

GO / NO-GO GATES

Gate 1 — mixing & continuity → NO-GO

Continuous (1 connected component, 100% of cells) BUT not mixed: across the integration-strength sweep 73%→56%→1% of cells sit in patient-pure Leiden clusters (θ=1/2/3); diff-sample neighbour fraction 0.30 at θ=2 vs 0.69 ceiling. The purest clusters (99% P11) are non-cycling genuine patient-private CNV states, not a shared axis. P11+P01 alone = 72% of cells. θ=3 forces mixing only by over-integrating (a detached speck appears). Do not force a trajectory.

Gate 2 — module geography → NA

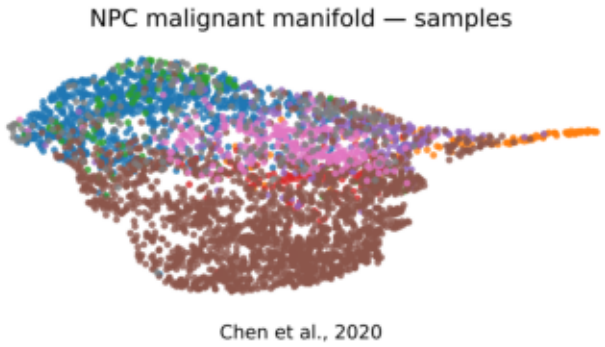
Not evaluated — gate chain stops at Gate 1 NO-GO (no manifold / signatures / correlation / modules notebooks produced).

Gate 3 — cross-repression → NA

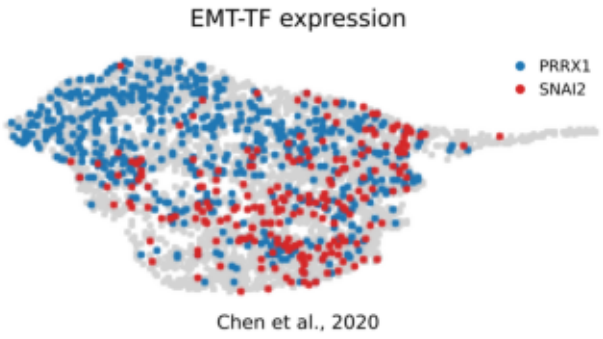
Not evaluated — no manifold, no hub.

STEP FIGURES

Manifold — samples



EMT-TF expression



Dataset.
EBV-associated nasopharyngeal carcinoma (Chen et al., 2020), 10x; 3,674 malignant cells across 8 samples.

Verdict.
Pattern does NOT reproduce — Gate 1 NO-GO. The malignant compartment clusters by patient, not into a sample-integrated continuous manifold, so no trajectory is fit.

Parameter choices

Min-cell threshold	62 (median-restricted spillover gap 107→36, ×2.97)	**Overdispersed genes**	1,290 (GAM)
n_pcs	6 (scree elbow)	**Integration**	Harmony θ=2 (balance of the θ=1–3 sweep)
Geometry	none — Gate 1 NO-GO, no trajectory	**EMT-TFs (descriptive, ≥3%)**	PRRX1 62% · SNAI2 24% · FOXC1 13% · ZEB2 13% · ZEB1 7% · TWIST2 6%; SNAI1/TWIST1 below floor